International and Monetary Economics

Course Code: 30054

Professor: Dmitriy Sergeyev

Lecture 6

Announcements

- **Quiz 1**
  - March 2, 2026 (end of Lecture 7)
- **Office hours**
  - Tuesday, 17:15-18:45, Office: 5-e1-02

Roadmap

1. Money and Interest Rates
   - Money demand
   - Money market equilibrium

2. Money and FX markets equilibrium (in the short run)

Money Demand

Individual Money Demand

- **Interest rates/expected rates of return**
  - (Usually) no interest on money
  - R = opportunity cost of holding money (interest rate on gov. bonds)
- **Risk**
  - The risk comes from unexpected inflation, which reduces the purchasing power of money.
  - But the same risk applies to other nominally safe assets (e.g., bonds)!
- **Liquidity**
  - The fact that money is a convenient medium of transaction is the key reason why people want to hold it despite a lower return compared to government bonds
  - The volume of transactions will affect the demand for money

Aggregate Money Demands

- Aggregate money demand is the sum of individual money demands of people, financial and non-financial firms

- Determinants
  - Interest rates (opportunity cost)
  - Income
  - Price level (because money is usually a nominal variable)

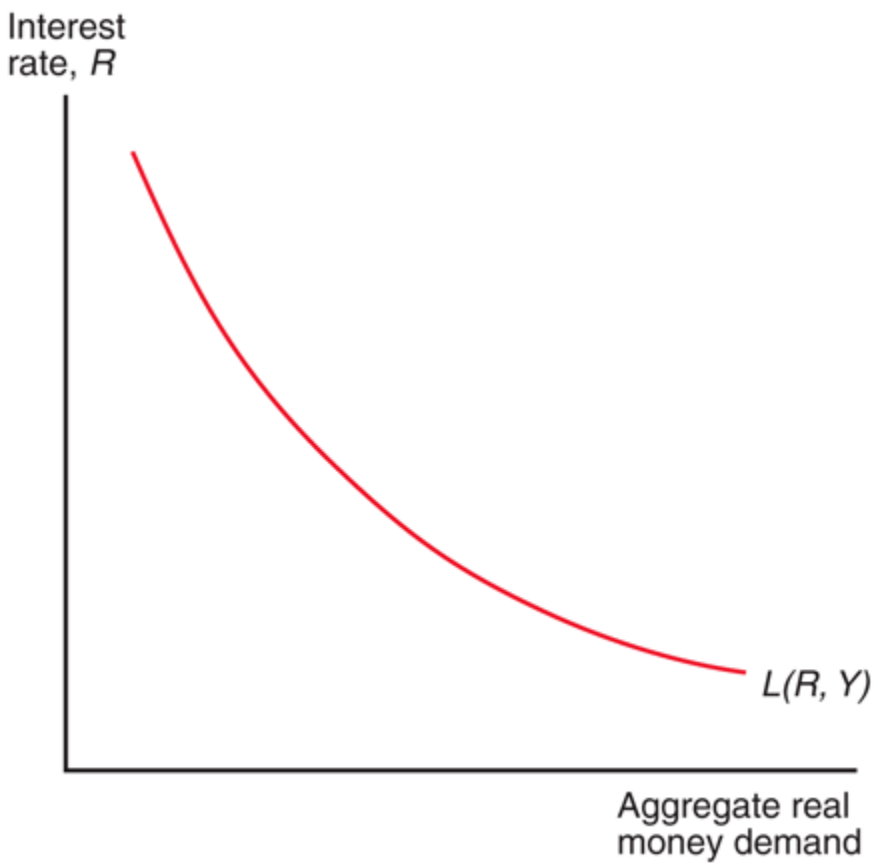
Aggregate Money Demand

- The aggregate demand for money can be expressed as:

$$\frac{M^d}{P} = L(R, Y)$$

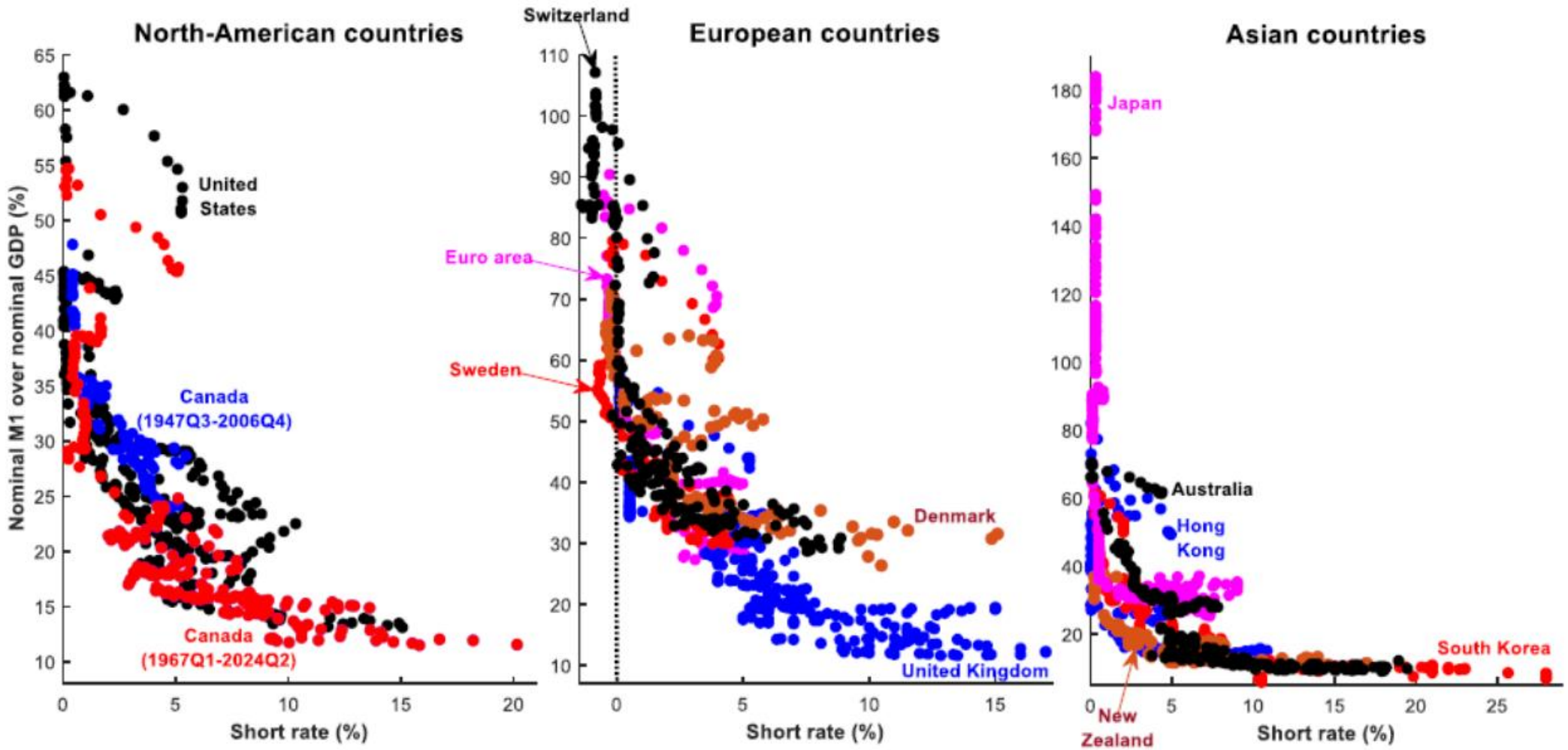
  - P is the price level
  - Y is real national income
  - R is a measure of interest rates on nonmonetary assets
  - L(R,Y) is the aggregate real money demand

Aggregate Money Demand



What happens to the demand for money when the interest rate falls to zero?

M1/nGDP vs. the Short Rate



Correlation is not causation!!!

Source: Benati, Nicolini (2025)

M1/nGDP vs the Short Rate in the US

Breakdown in Money demand in 1980?

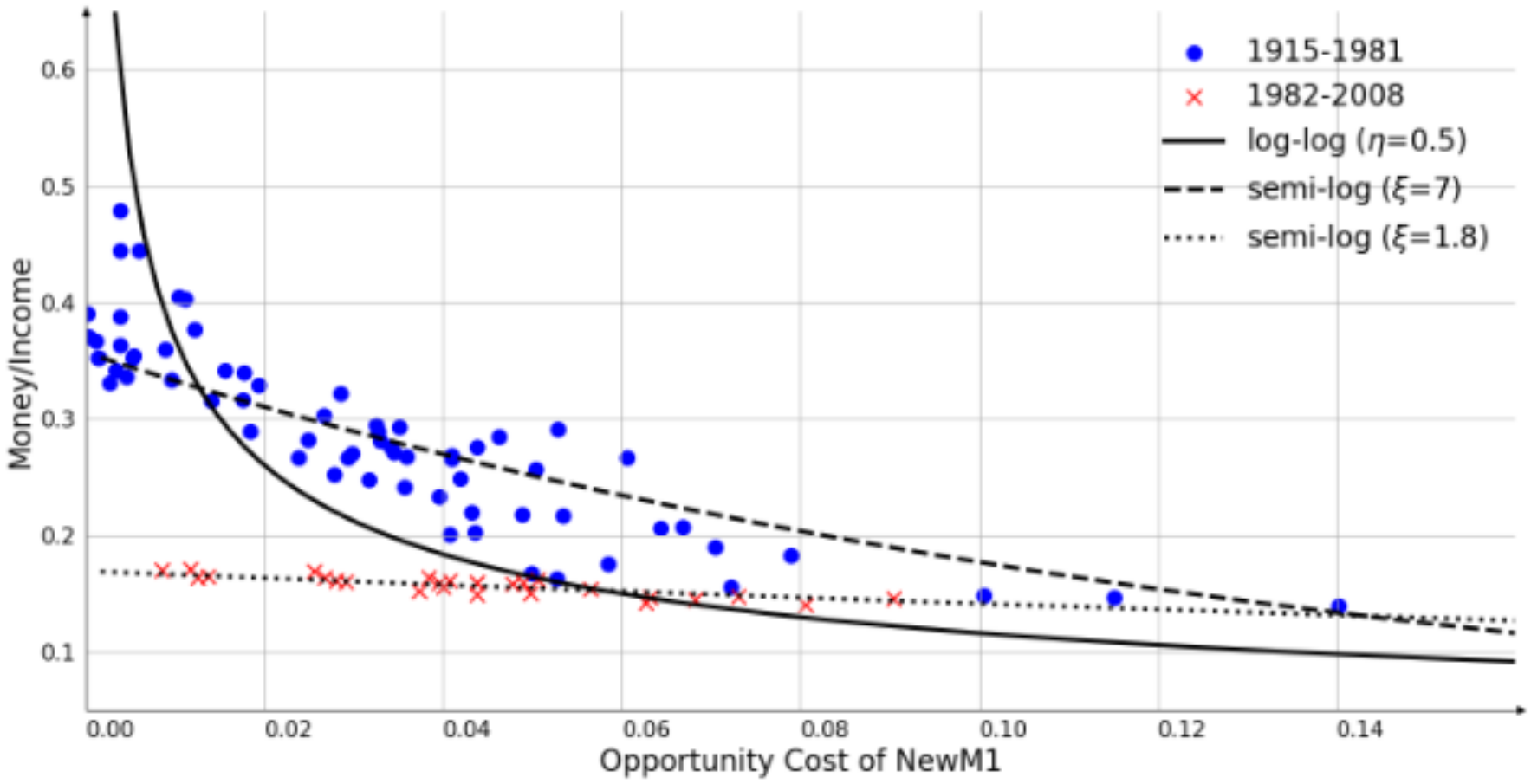
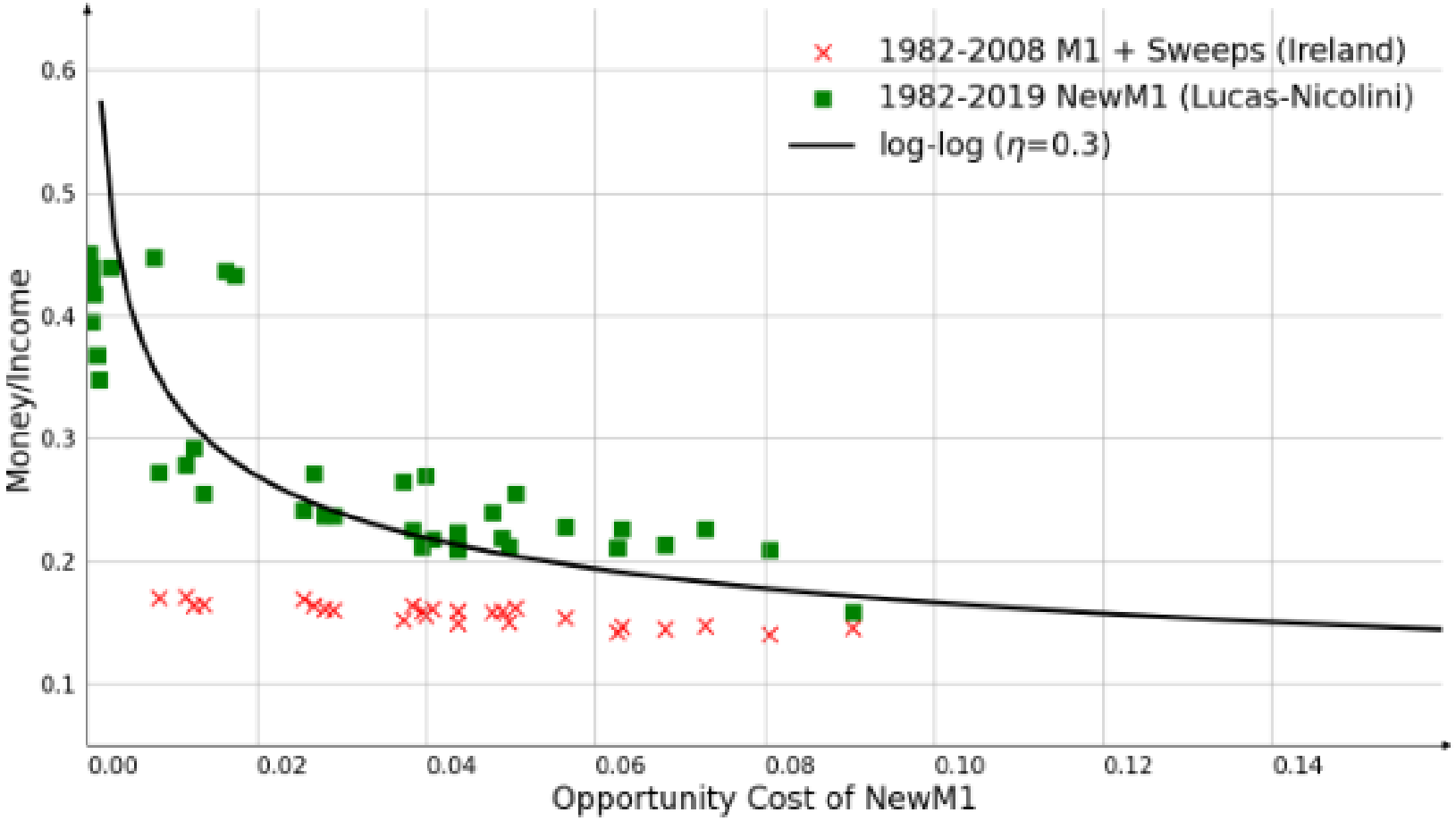


Figure 6: US real money demand, 1915-2008

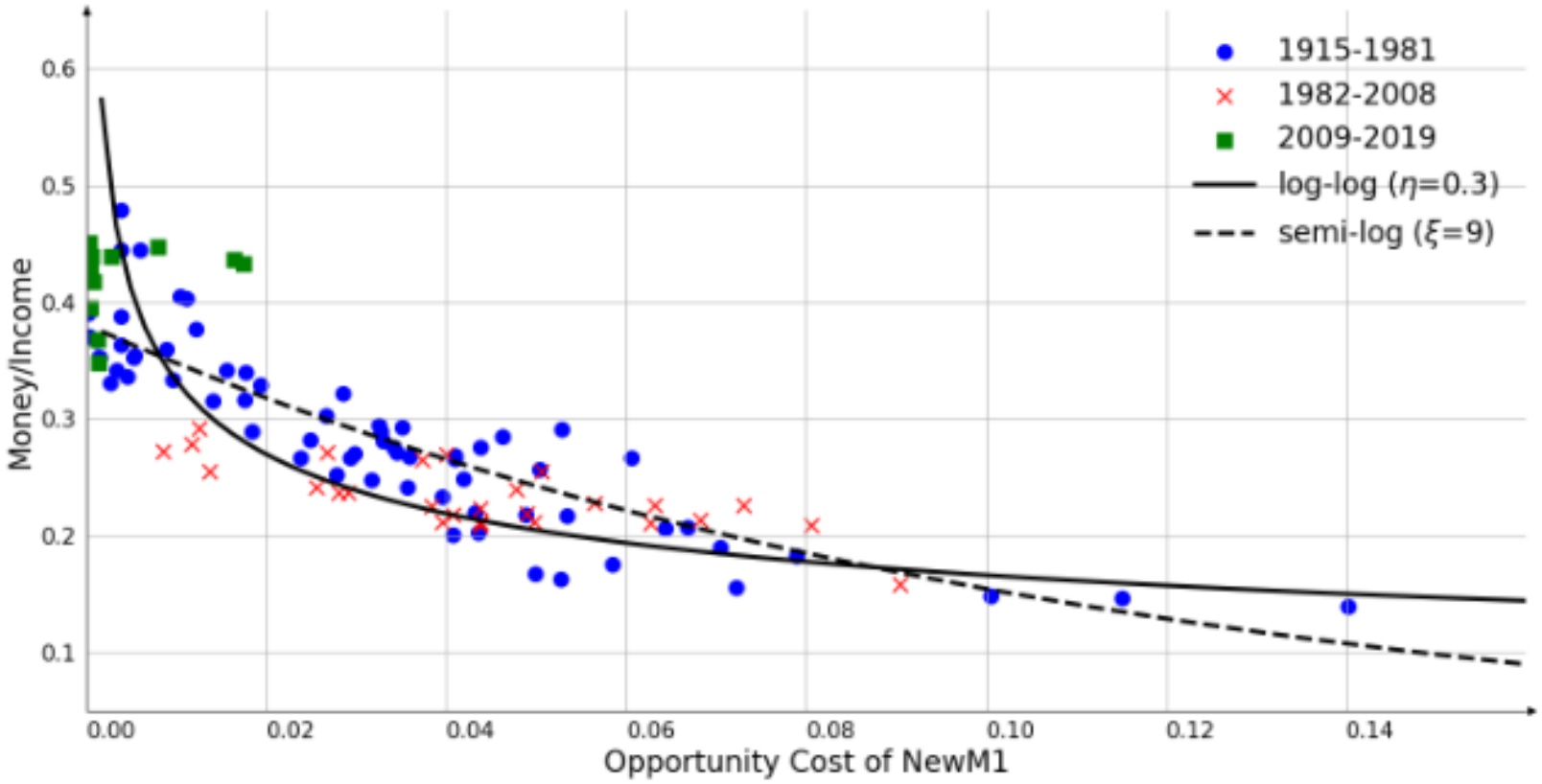
The Measure of Money Matters



Sweeps: checking accounts transferred to Money Market Demand Accounts during the night (MMDA)

NewM1 = M1 + MMDA

NewM1/nGDP vs the Short Rate in the US



Log-log money demand:

$$\ln \frac{M_t}{P_t y_t} = a^1 - \eta \ln r_t + u_t^1$$

Semi-log money demand:

$$\ln \frac{M_t}{P_t y_t} = a^2 - \gamma r_t + u_t^2$$

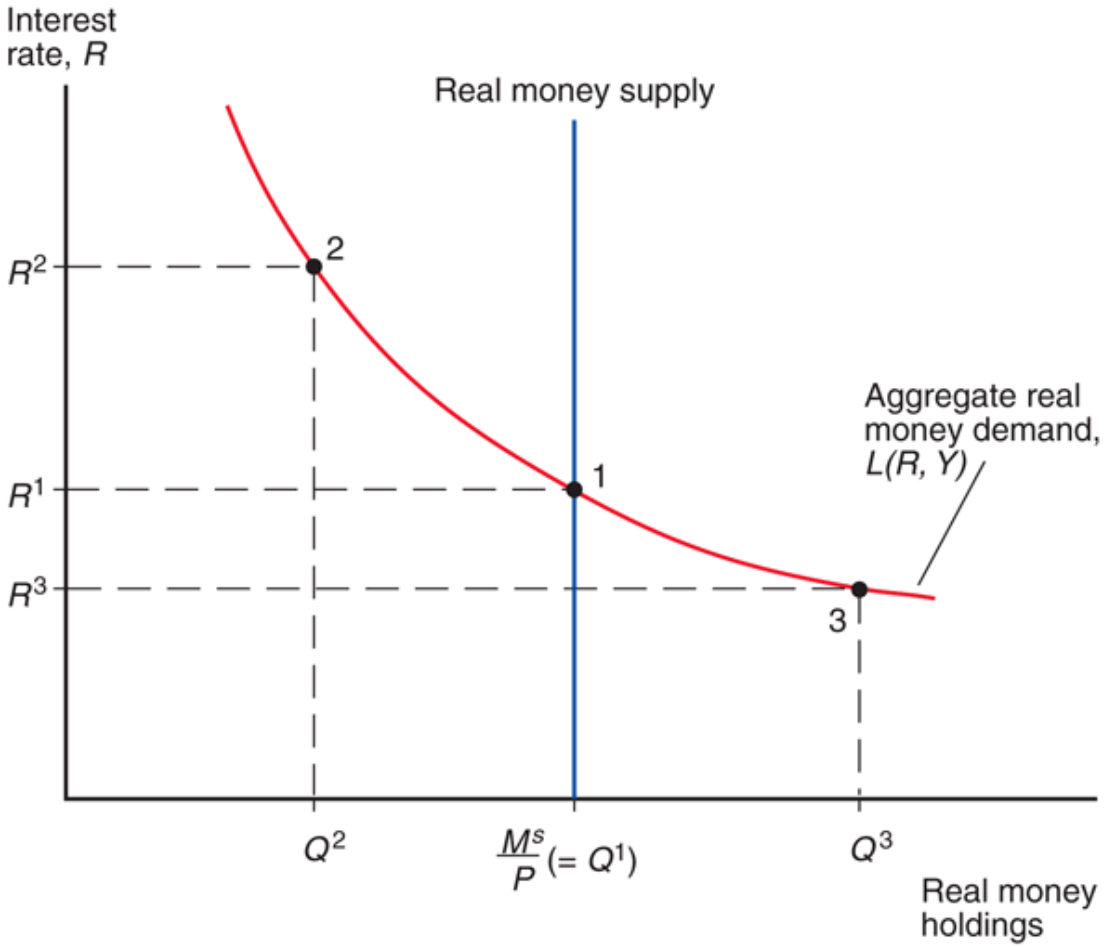
Money Market Equilibrium

Money Market Equilibrium

$$\frac{M^s}{P} = \frac{M^d}{P}$$

$$\frac{M^s}{P} = L(R, Y)$$

Money Market Equilibrium



Money Market Equilibrium

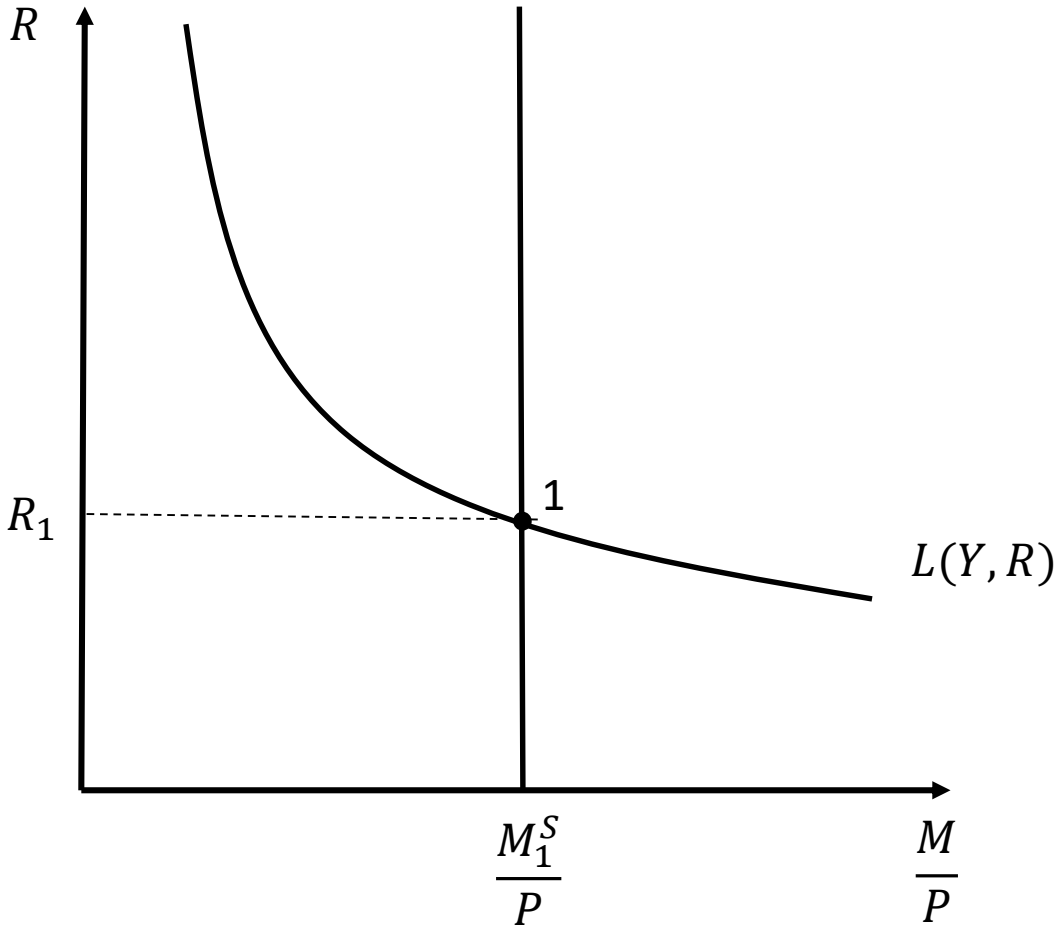
- Adjustment to equilibrium
  - Suppose $M^{\mathrm{s}} > M^{\mathrm{d}}$
  - People are willing to buy interest bearing assets (bonds)
  - The price of these assets increases

    (or the interest rate decreases)
  - Money demand goes up

$$M^s = M^d$$

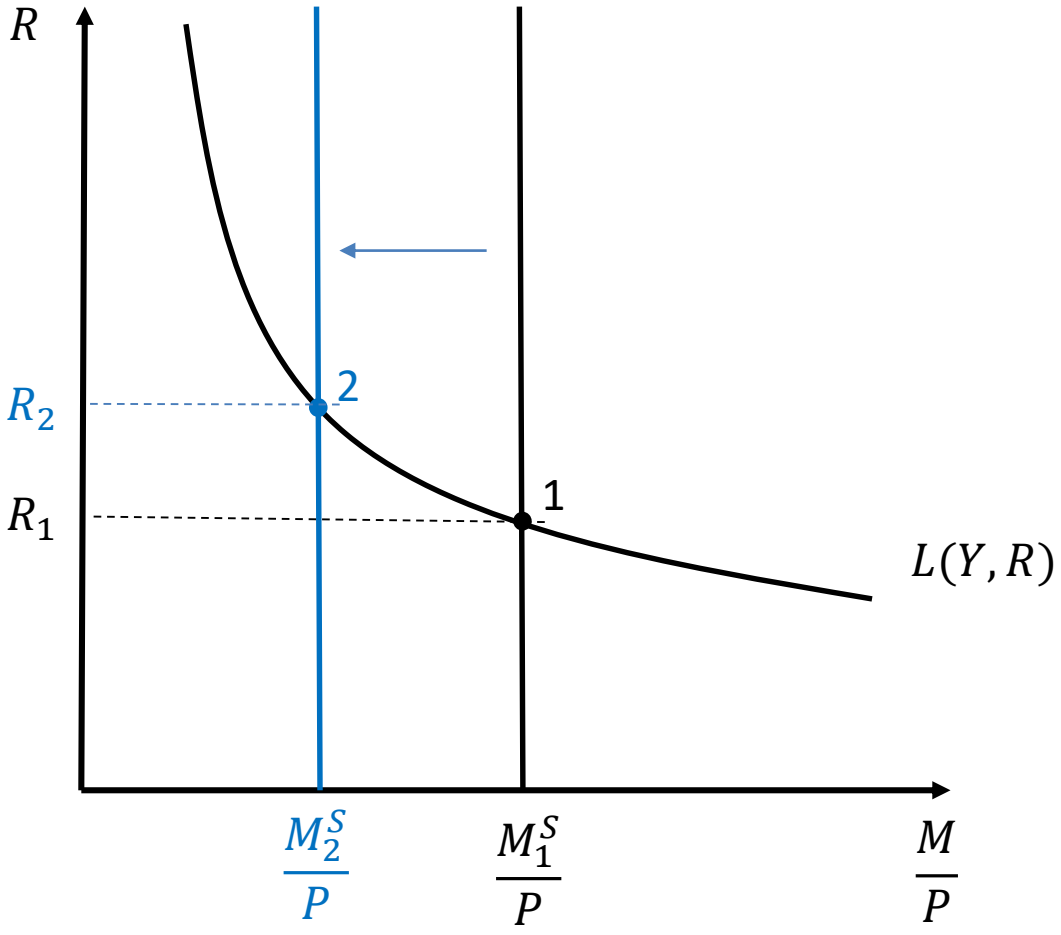
Exercise 1

A Decrease in Money Supply



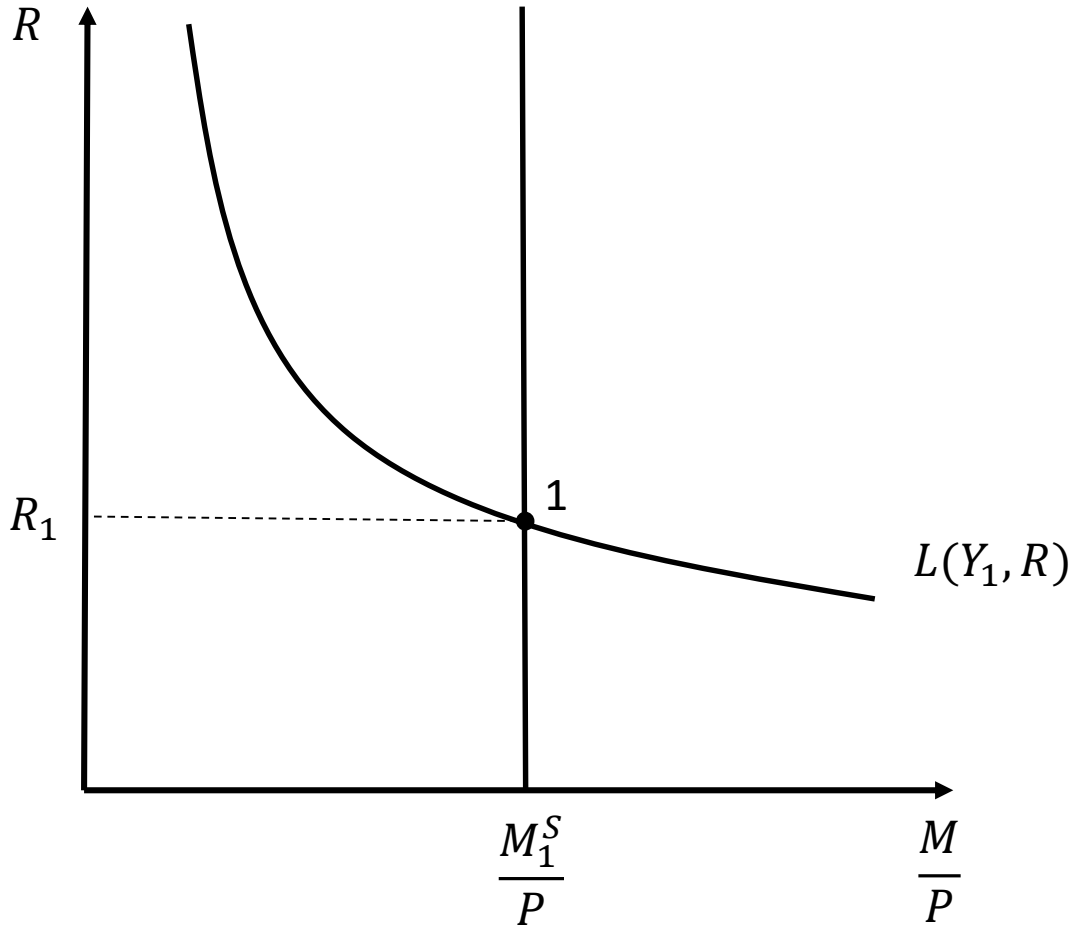
Exercise 1

A Decrease in Money Supply



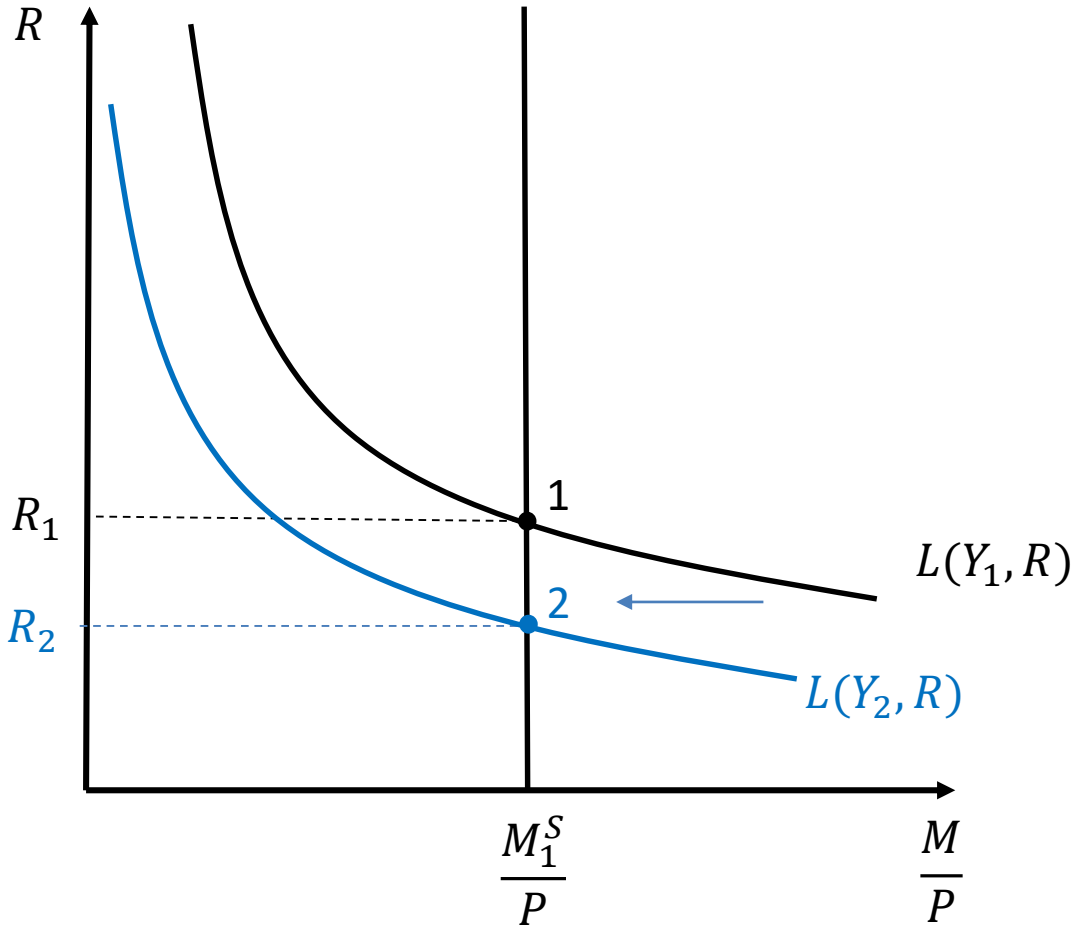
Exercise 2

A decrease in real income



Exercise 2

A decrease in real income

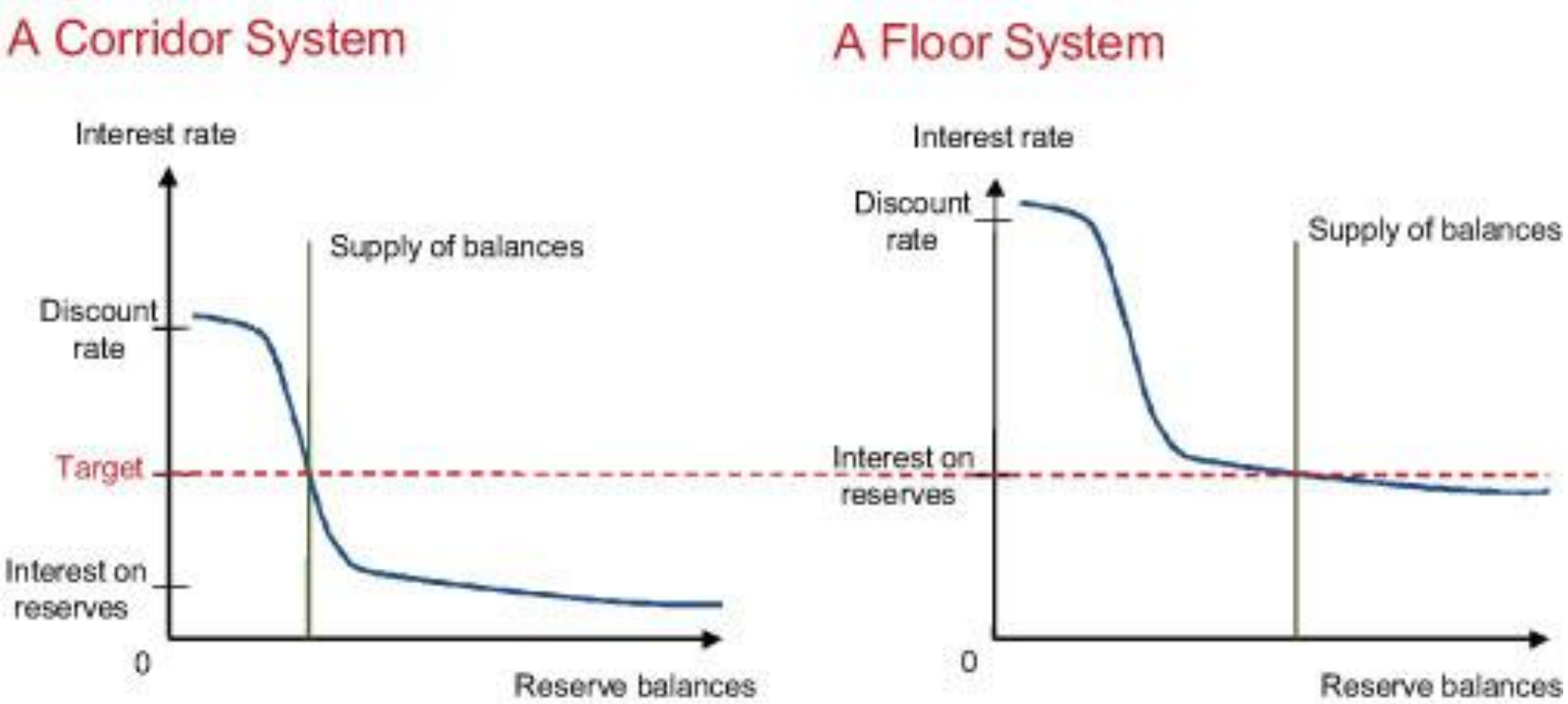


Monetary Policy in Reality

- **Modern monetary policy is, in fact, an interest rate policy**
  - Money demand is unstable, so targeting the money supply leads to variations in the interest rate
  - These days, most central banks announce the target interest rate and adjust their policy tools (e.g., money supply or interest rate tools)

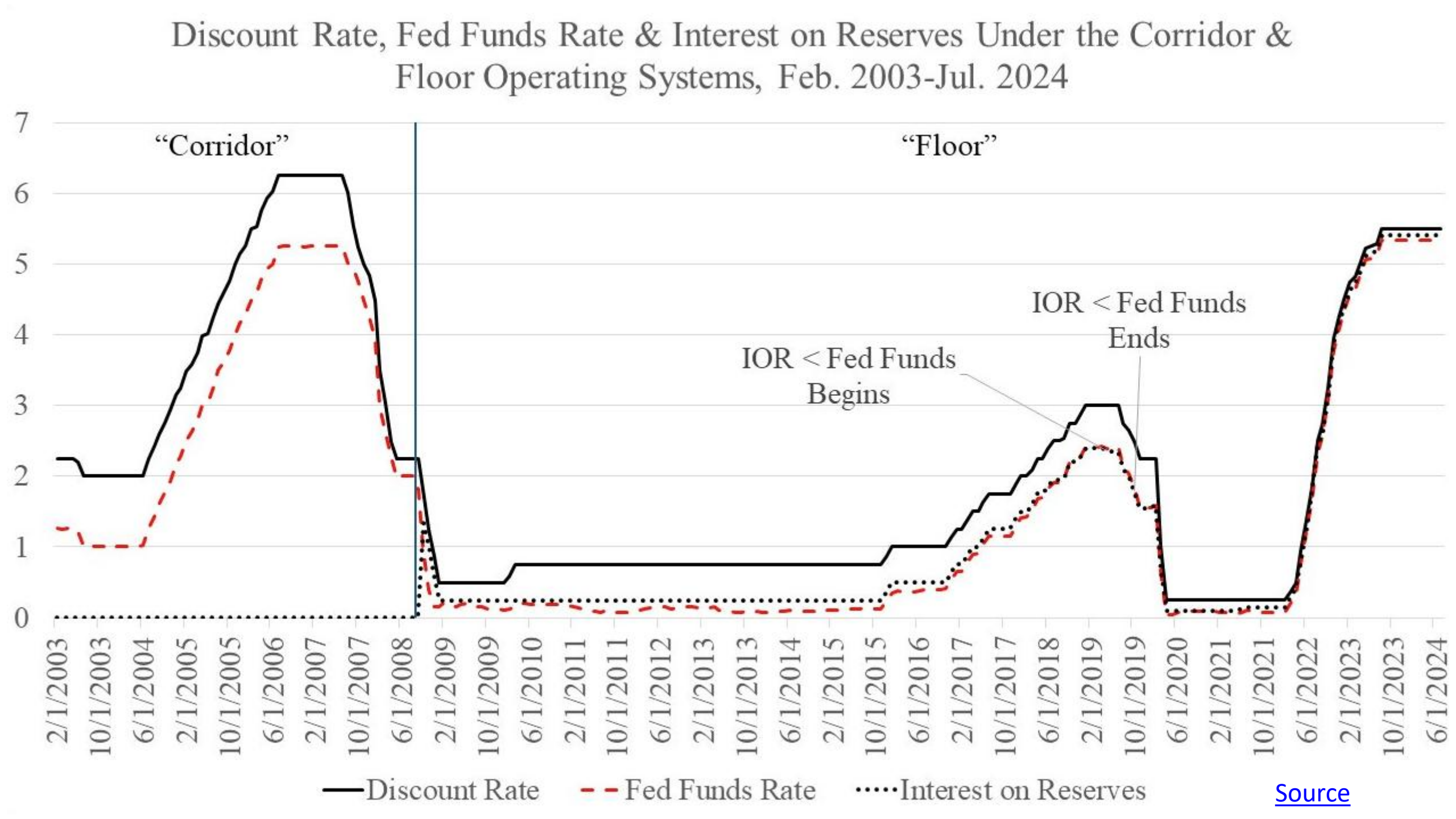
- **The Global Financial Crisis made central banks issue enormous amounts of reserves to provide liquidity to the financial sector**
  - Need additional (interest rate) tools if they want to control interest rates

Corridor vs. Floor System



Source: Federal Reserve Bank of New York

The Federal Reserve influences the target interest rate in the federal funds market—the market where banks lend federal reserve balances to each other—through three main tools: the discount rate, the interest on reserves, and the supply of federal reserve balances.


Discount Rate, Fed Funds Rate & Interest on Reserves Under the Corridor & Floor Operating Systems, Feb. 2003-Jul. 2024
"Corridor"
"Floor"
IOR < Fed Funds Begins
IOR < Fed Funds Ends
7
6
5
4
3
2
1
0
2/1/2003
10/1/2003
6/1/2004
2/1/2005
10/1/2005
6/1/2006
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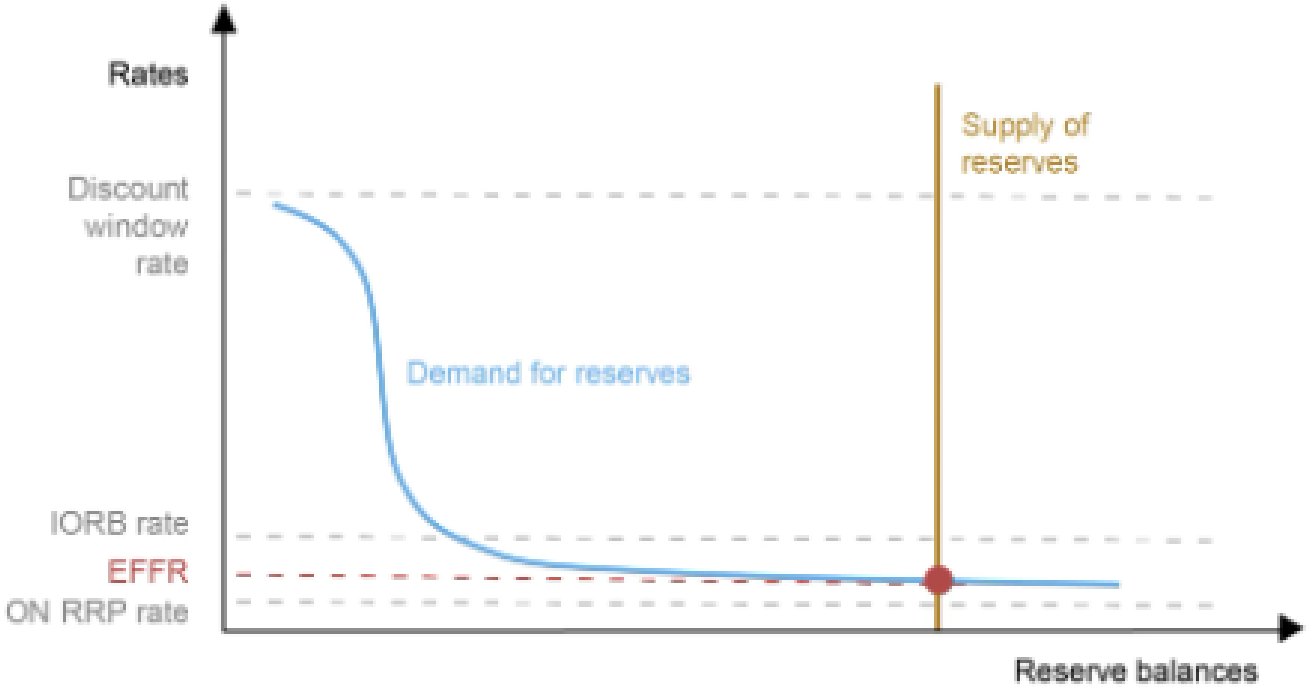
Discount Rate
Fed Funds Rate
Interest on Reserves

Source

The Federal Reserve

Banking System Reserve Demand Curve

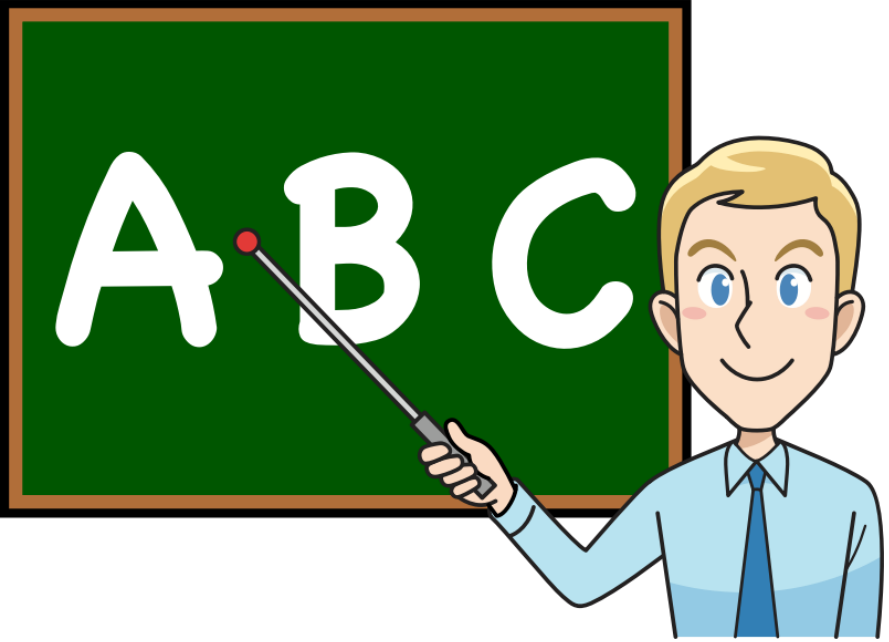


Notes: The discount window rate is the interest rate charged to depository institutions on loans they receive from the discount window facility. The overnight reverse repurchase agreement (ON RRP) rate is the maximum interest rate that the Federal Reserve is willing to pay at the ON RRP facility. IORB is interest on reserve balances. The effective federal funds rate (EFFR) is represented by the red circle, where the demand for reserves and the supply of reserves intersect.

- IORB rate – interest rate on reserve balances (available to only depository institutions)
- EFFR – effective federal funds rate
- ON RRP rate – overnight reverse repurchase agreement rate (used by non-depository institution, such as money market mutual funds)

Source: Federal Reserve Bank of New York

Money and FX Markets Equilibrium (in the short run)



English Language

- **Short-run** and **long-run** are <u>adjectives</u>
  - Example: "short-run effects of a contractionary monetary policy"
- **Short run** and **long run** (i.e., without a hyphen) are <u>nouns</u>
  - Example: "In economics, the long run refers to a theoretical state where all markets have reached equilibrium."

Money and FX Markets Equilibrium

- FX market

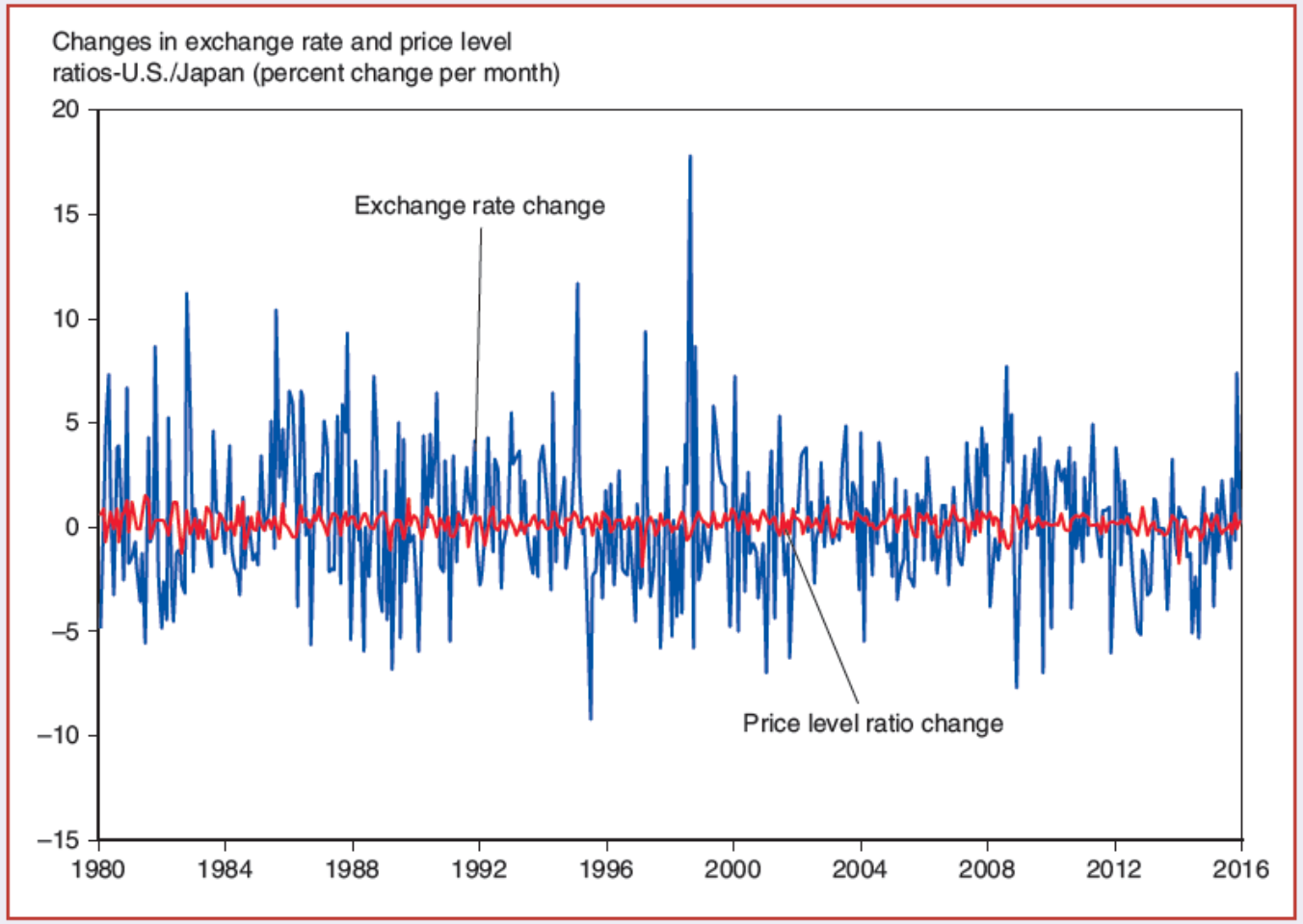
$$R = R^* + \frac{E^e - E}{E}$$

- Money market (at home)

$$\frac{M^s}{P} = L(R, Y)$$

To proceed we need to know how aggregate output (Y), the price level (P), and the future exchange rate (E^e) are determined!

- Assumptions
  1. Y, Y^* are fixed **for simplicity**
  2. E^e is fixed **for simplicity** (because perhaps policy experiments are ***temporary*** and do not affect future exchange rate)
  3. P, P^* are fixed **in the short run**

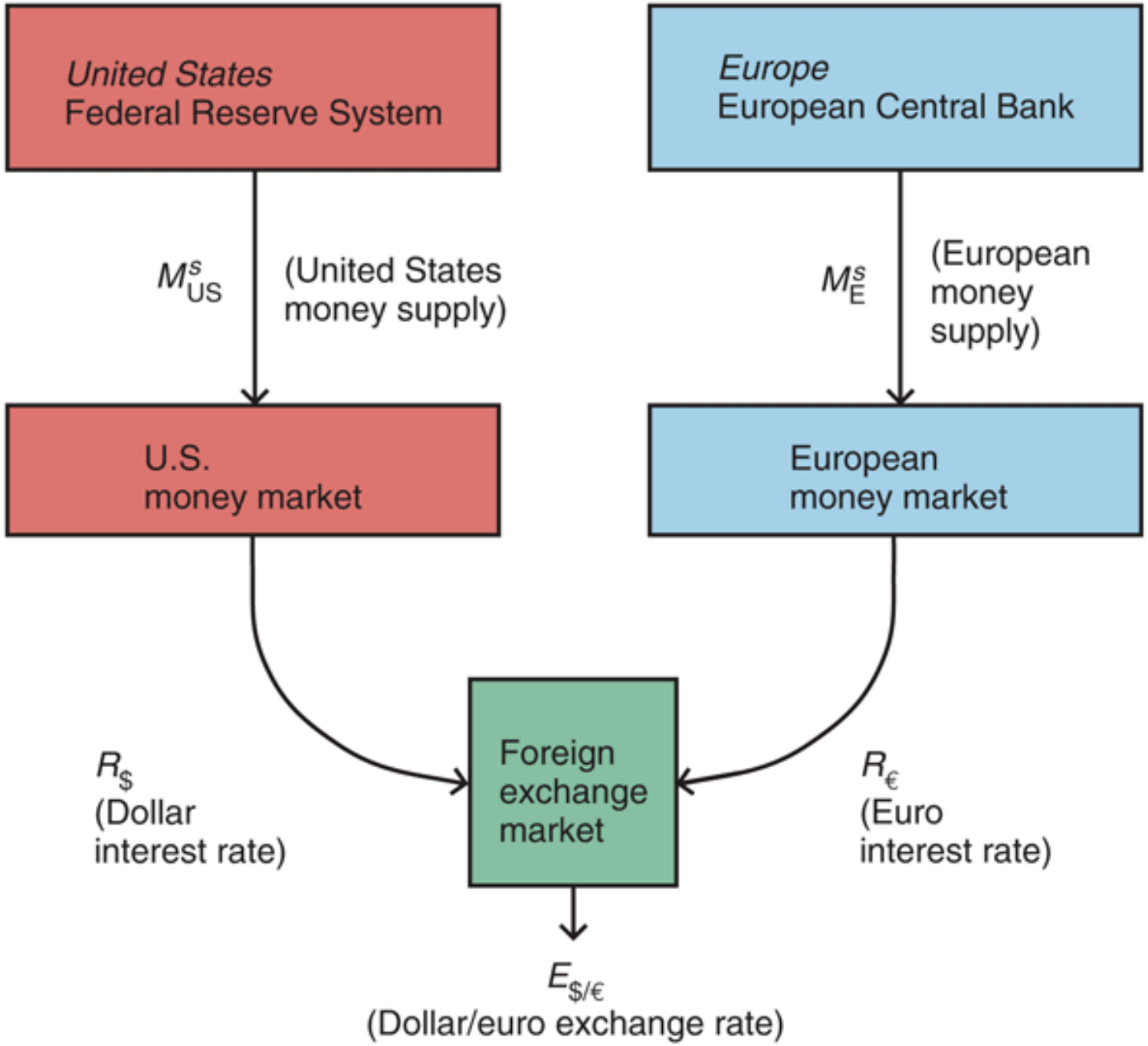


Pearson MyLab Economics Real-time data

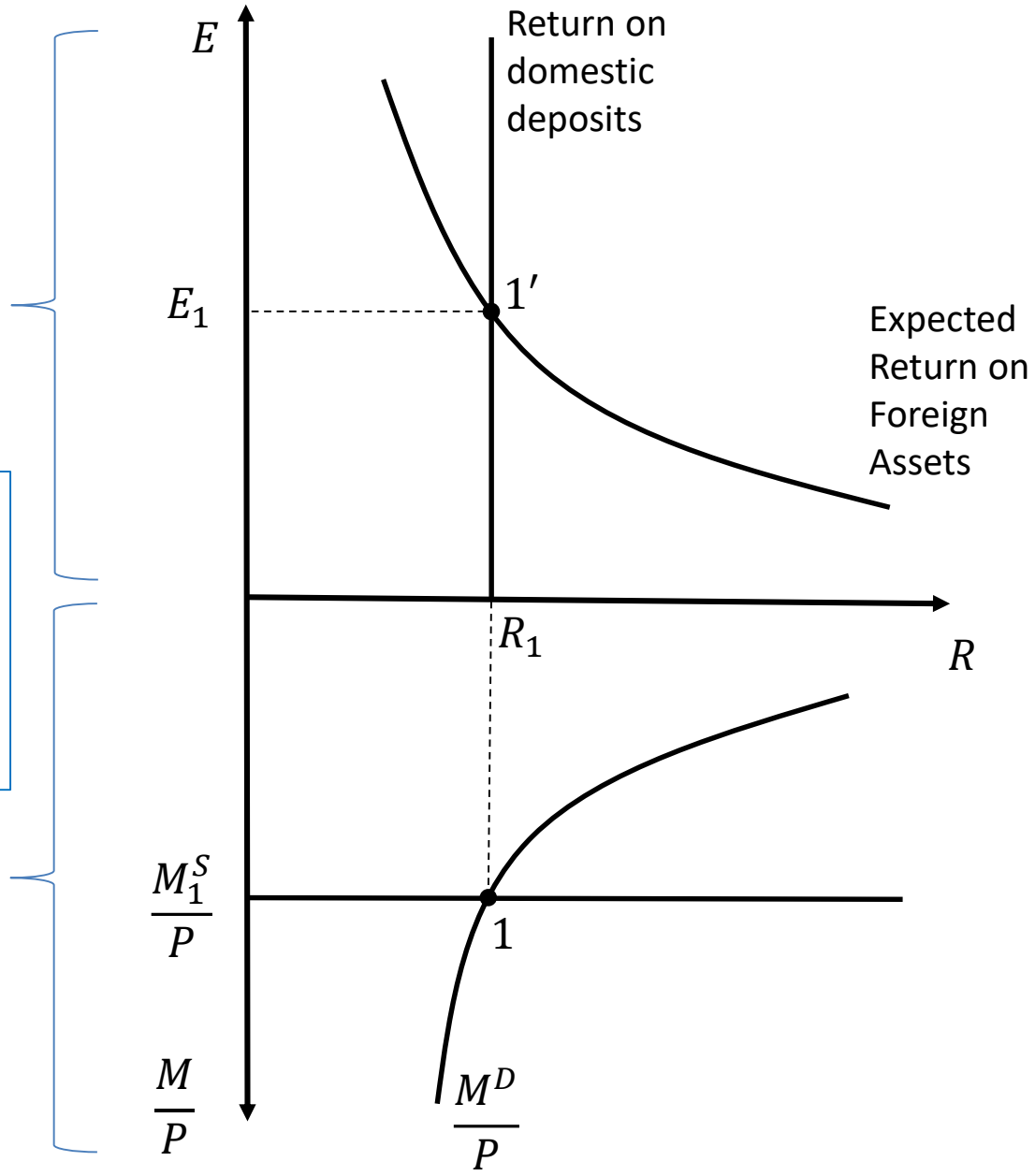
FIGURE 15-11

Month-to-Month Variability of the Dollar/Yen Exchange Rate and of the U.S./Japan Price Level Ratio, 1980–2016

- In the short run, the nominal exchange rate is much more variable than the price level
- **But** this may not always be the case
  - Example: price indexation during hyperinflations


United States
Federal Reserve System
Europe
European Central Bank
M^s_{US}
(United States money supply)
M^s_E
(European money supply)
U.S. money market
European money market
Foreign exchange market
$R_\$$
(Dollar interest rate)
$R_€$
(Euro interest rate)
$E_{\$/€}$
(Dollar/euro exchange rate)

Simultaneous Equilibrium in the home Money and the FX Markets



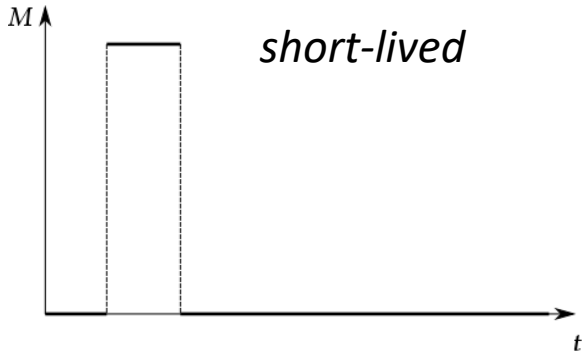
Exercise 1

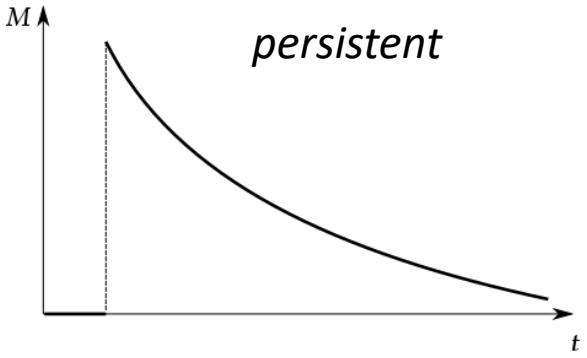
An Increase in domestic money supply

- How does the current exchange rate change when a domestic central bank **temporarily** increases its money supply?

Language

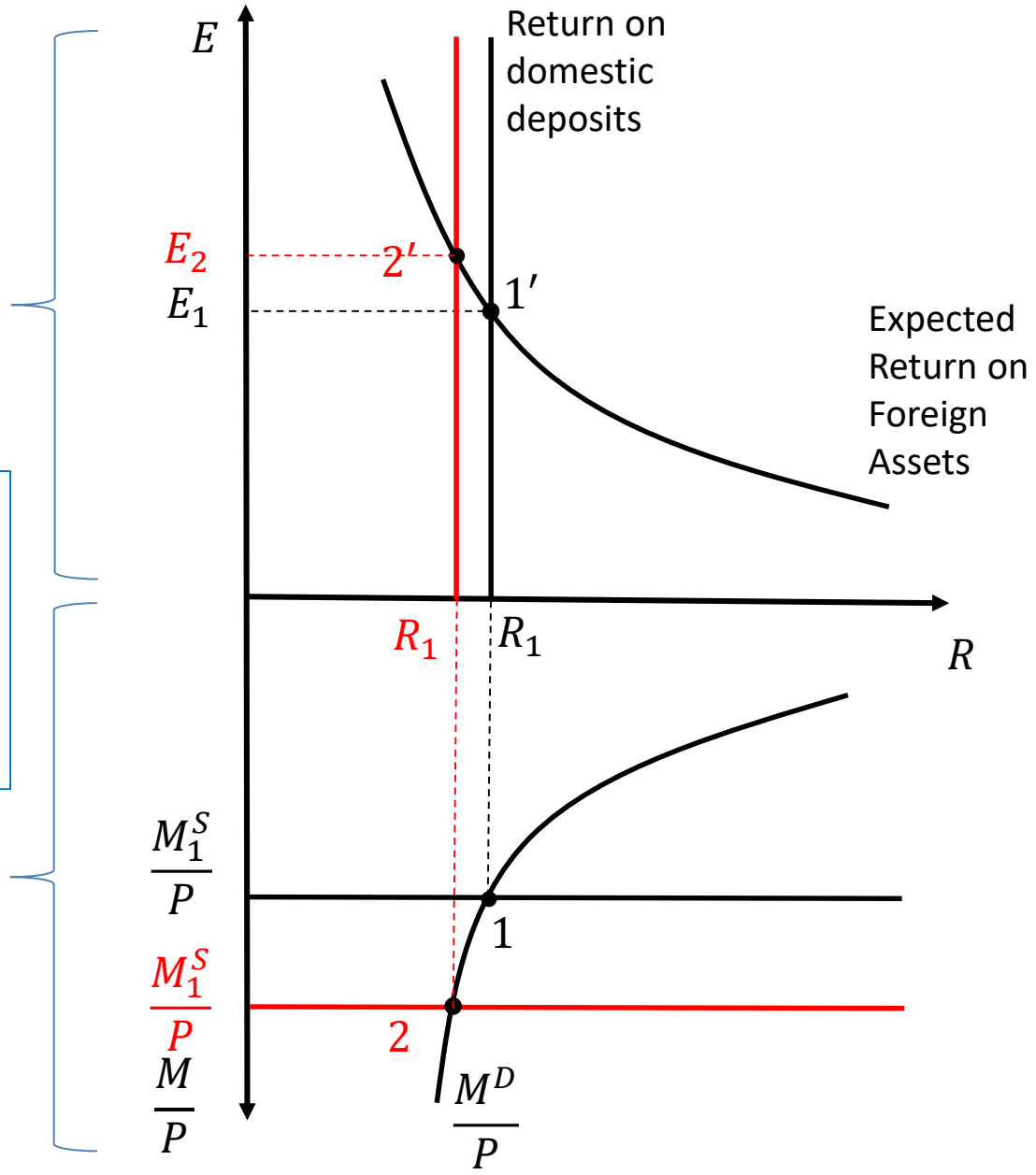
- A temporary change





- A permanent change

Simultaneous Equilibrium in the home Money and the FX Markets



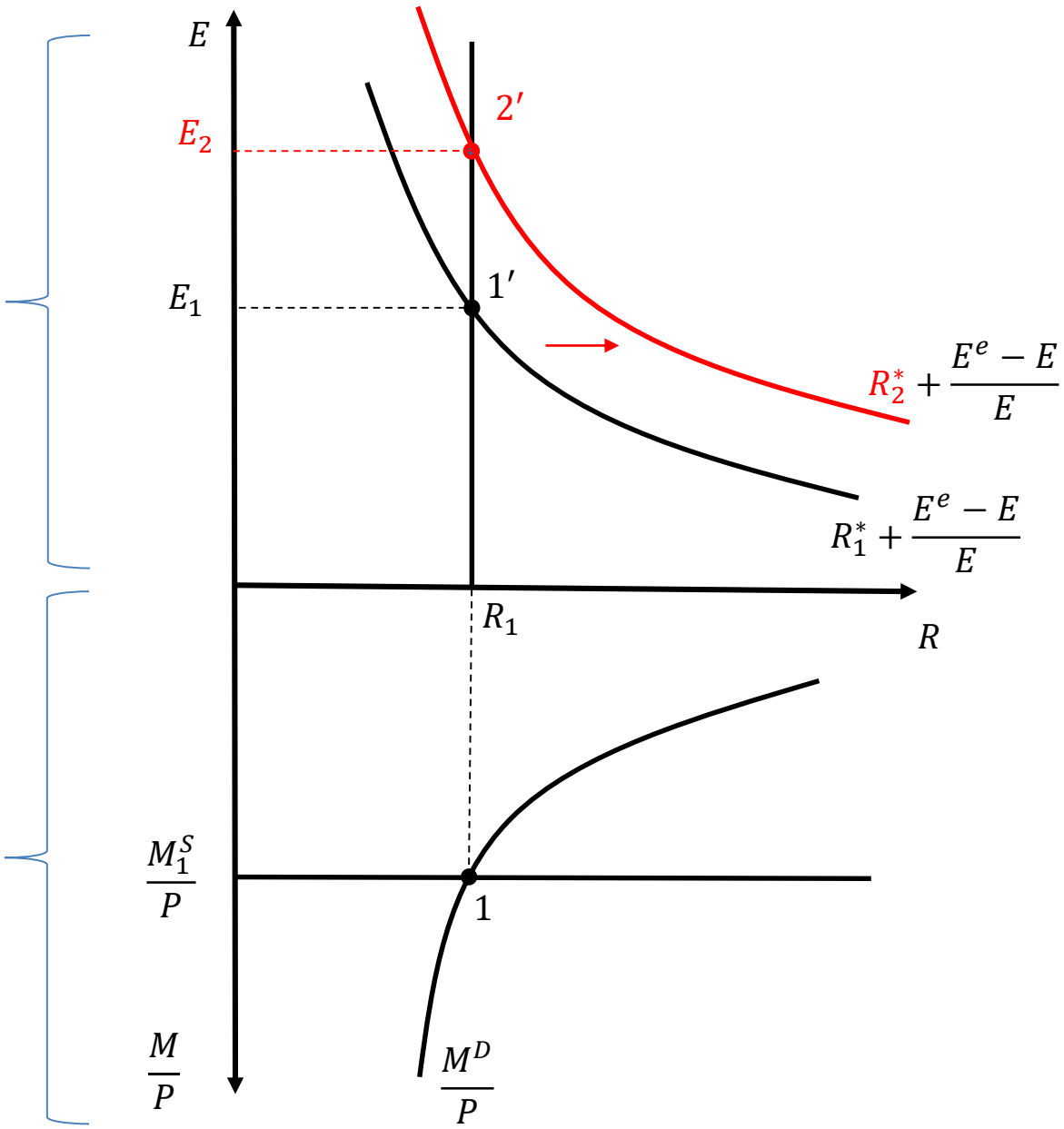
Exercise 2

A Decrease in foreign money supply

- How does the current exchange rate change when foreign money supply shrinks temporarily?

Change in foreign monetary policy

$$M^{*S} \downarrow \Rightarrow R^* \uparrow$$



Readings

- Next lecture:
  - KOM, Chapter 16